

Quantum Optimization for the Vehicle Routing Problem (VRP)

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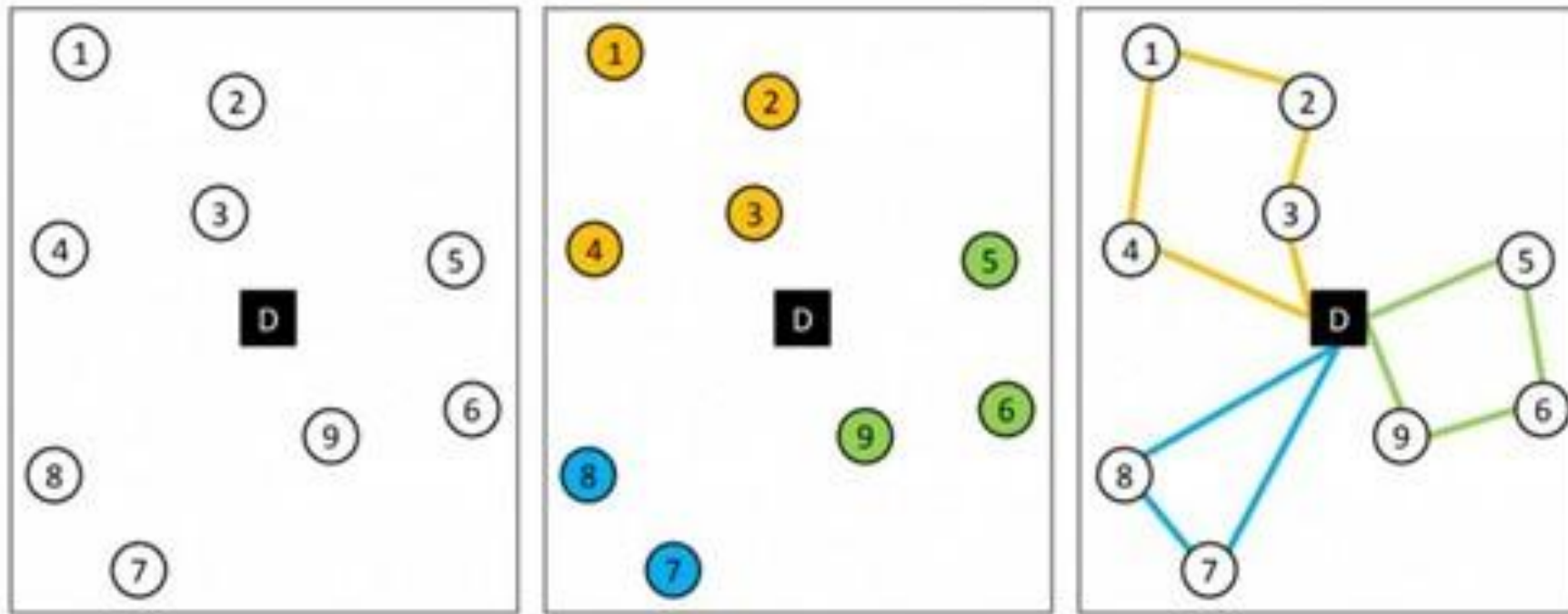
Supervisor: [Udaya Parampalli](#)

May 19th 2025

Work in Progress Oral Presentation

Quick Illustration

Vehicle Routing Problem (VRP) is a concept in the field of transportation that involves identifying the most efficient route for a set of customers with fixed positions (Khan, Alghamdi, & Khan, 2023) [*TSP derived problem*]



Unassigned Customers

Customer Segmentation

Optimal Paths

NP-Hard!
 $T(n) = \Omega(n!)$

Quantum Approximate Optimisation Algorithm (QAOA)

QAOA: Quantum (more accurately hybrid quantum-classical) algorithm that aims to approximate solutions to combinatorial optimization problems. (Farhi et al., 2014)

QAOA can be prepared by defining:

- **Qubits** (represent the problem and output the solution)
- **Hamiltonian and Cost Function** with trainable parameters (as an objective function)
- **Quantum Circuits** (that we will use to iterate through the parameters)

This research will also utilize:

- Reducing QAOA qubits for small-scale quantum machines (Leonidas et al., 2023)
- Quantum optimization strategy specifically for vehicle constraints (Fitzek et al., 2024)

Objectives

1. Design Optimization Techniques for VRP using Quantum Algorithms [RQ1, RQ2]
2. Explain why VRP solution based on quantum approaches may perform better than classical approaches [RQ3]

Research Questions

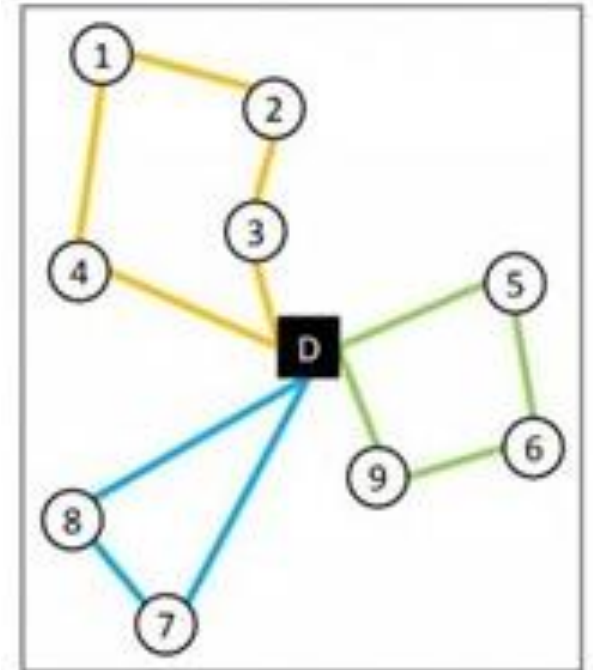
- RQ1: How can the spatial data of VRP be converted and formulated into a structure that is solvable by quantum algorithms?
- RQ2: How can we allocate qubits to solve the VRP using QAOA?
- RQ3: To what extent are quantum algorithms efficient and scalable compared to best-performing classical VRP solvers?

Case Study: Waste Collection in City of Casey

- **Depots**: Waste collection operations points (data.casey.vic.gov.au)
- **Customers**: Public litter bins' location (data.casey.vic.gov.au)
- **Routes**: Victoria's road network (vicdata.vicroads.vic.gov.au)

Constraints

- Traffic level
(vicroadsopendata-vicroadsmaps.opendata.arcgis.com)
- Weather conditions (data.casey.vic.gov.au)
- Road Crashes (discover.data.vic.gov.au)



Research Methodology: QAOA for VRP

- ✓ 1. Vehicle Routing Problem (VRP) and Graph Modeling
 - translate spatial data (city of Casey) into graph data structure
- ✓ 2. Qubit Allocation
 - Allocate qubits (big enough to capture VRP solution but small enough to not waste memory)
- ✓ 3. Cost Function and Encoding Scheme
 - Simple enough to be iteratable in circuits with a consideration in number of qubits
- 4. Hamiltonians and Quantum Circuits
 - preparing quantum circuits and hardware (quantum cloud) to run
- 5. Handling Constraints
 - Calculating constraints can be done prior to qubit allocation (dynamic)

Preliminary Results

Num of customers (n)	Number of depots (d)	Number of vehicle (v)	Theoretical Minimum Qubits Needed	Moderate Qubit Allocation	Ideal Qubits Needed (Naïve approaches)
5	1	1	10	22	44
9	1	3	27	65	116
9	2	3	30	67	150

City of Casey

d = 5

n = 889

Contribution

Qubit efficient way to solve the VRP

- Trade-off between number of qubit and time complexity
- As quantum computing hardware is costly

Encoding Method	Qubit Count	Num of Qubit	Cost Function Simplicity
Naive Matrix (NxN)	N^2	High	Simple
Logarithmic Permutation	$\lceil \log_2(n!) \rceil$	Low	Complex (impractical)
Node-Visit-Time	$n \cdot \lceil \log_2(N) \rceil$	Moderate	Moderate

Reduced Time Complexity

- n customers, d depots, v vehicles
- QAOA: s iterations, p depth
- At what point the $T(\text{quantum}) < T(\text{classical})$?

$$T(n, d, v)_{\text{Classical}} \approx (n + d)! \cdot v^n$$

$$T(n, d, v, p, s)_{\text{Quantum}} \approx s \cdot p \cdot v \cdot (n + d)^2$$

THANK YOU

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Timeline

[illegible]

Formulas

Encoding Method	Qubit Count	Num of Qubit	Cost Function Simplicity
Naive Matrix (NxN)	N^2	High	Simple
Logarithmic Permutation	$\lceil \log_2(n!) \rceil$	Low	Complex (impractical)
Node-Visit-Time	$n \cdot \lceil \log_2(N) \rceil$	Moderate	Moderate

$$Q_{\text{total}} = d + v + v \cdot d + v \cdot n$$

$$Q_{\text{total}} = \lceil \log_2 [(v + (d - 1))!] \rceil + \lceil \log_2 [(n + (v - 1))!] \rceil$$

$$Q_{\text{total}} = \lceil \log_2(d) \rceil + \lceil \log_2(v) \rceil + \lceil \log_2(d \cdot v) \rceil + \lceil \log_2(v \cdot n) \rceil$$